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AgriSuit: Pixel-Based Parcel Suitability Analysis

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1 Abstract

This study introduces AgriSuit, an R Shiny web dashboard that connects crop-suitability maps with real-estate listings to support quick and transparent agricultural property screening. The platform analyses official suitability data to identify the main environmental factors that limit growth and calculates the percentage of land suitable for each crop within a parcel. These results are summarised visually, helping users understand which crops are most viable and why. Through its interactive dashboard, users can explore parcel-level suitability maps, view soil and climate details, and open property listings directly to assist in due-diligence and investment decisions. In pilot use, AgriSuit delivered clear and accessible results for non-technical users while revealing current limitations, such as a fixed list of crops and simplified climate assumptions. Future development will focus on adding climate scenario options, uncertainty indicators, and water-access and economic layers to make the tool more comprehensive and scalable across regions.

2 Introduction and Motivation

The Tasmanian Enterprise Suitability Map (TESM) is a geospatial decision-support tool that integrates soil, climate, and landscape data to classify land as well suited, suitable, marginally suitable, or unsuitable for a range of agricultural enterprises. Developed by the Tasmanian Institute of Agriculture (TIA) in collaboration with industry, TESM supports planning and investment by providing sciencebased assessments of crop potential across Tasmania. By making suitability information publicly accessible through LISTmap, it has become a cornerstone of agricultural planning and resource allocation in the state.

Despite its strengths, existing platforms present two key limitations. First, they respond poorly to changes in agribusiness trends, including the emergence of new crops that may become economically viable or strategically important for food security in the near future. Second, they are not directly linked with the property market, leaving a critical gap between geospatial crop suitability analysis and the practical realities of land acquisition. Prospective buyers or agribusiness investors seeking to identify land parcels that align with their crop preferences must manually consult multiple sources, including property listing sites, agronomy reports, and local council data. This process is time-consuming, fragmented, and often discourages comprehensive due diligence.

Prospective buyers and landholders in Tasmania therefore lack an efficient mechanism to assess which crops are suitable for newly listed properties at the time they appear on the market. Current real-estate platforms do not integrate geospatial factors such as soils, climate, topography, or water access into actionable summaries, meaning that opportunities are frequently missed. This project

addresses that gap by automatically assessing crop suitability for new property sales listings and directly linking users with properties that match their preferred crops. To achieve this, the project develops an integrated system that combines property listing information, cadastral parcel boundaries, and statewide crop suitability layers. This system computes parcel-level suitability and presents the results through an interactive Shiny dashboard, providing users with a more personalised and efficient way to evaluate investment opportunities.

Building on this problem statement, the present study redefines the role of TESM by moving beyond the static functions of LISTmap. The project develops an integrated system that links live property sales data with crop suitability assessments derived from TESM, providing users with a more personalised and efficient way to identify land that matches their preferred crops. By combining property listing information, cadastral parcel boundaries, and statewide crop suitability layers, the system computes parcel-level suitability and delivers a more dynamic and actionable decision-support tool.

The significance of this research extends across several domains. For investors, the system enables better decision-making by reducing business risk and enhancing opportunity discovery. It assists those who already have specific crop strategies by identifying parcels that match their requirements as soon as they are listed, while also supporting landholders or capital investors who wish to explore which crops are most suitable for a given parcel. For the agribusiness sector, it provides a scalable framework for integrating suitability analysis into real-time land acquisition decisions, enabling faster and more confident responses in an environment of rising land values, shifting crop demands, and climate uncertainty. By reducing the risk of costly misallocation, the system ensures that capital and resources are directed toward opportunities with the greatest long-term viability. For policy and planning bodies, the system demonstrates how public geospatial resources such as TESM can be re-purposed into practical, user-focused applications that bridge information gaps between government datasets and private-sector decision-making. It also highlights that these resources are not confined to a small niche of agricultural specialists, but can deliver value to mainstream business communities, including property-based investors and developers. This broader relevance strengthens the case for continued improvement of TESM and LISTmap through the inclusion of additional variables and more sophisticated modelling approaches. By connecting suitability assessment with property market dynamics, this project contributes to the creation of a more responsive, transparent, and data-driven agricultural investment environment in Tasmania.

3 Objective

This project aims to integrate TЕСM crop suitability layers with live property listing data in order to generate parcel-level suitability assessments. Building on this foundation, it develops an interactive Shiny dashboard that allows users to view ranked property recommendations, examine limiting factors, and directly access the corresponding property listings. The project also evaluates the tool's functionality through case studies in Tasmania and explores its potential for broader application, extending to regions with comparable suitability datasets and to future industry-facing implementations.

4 Literature Review

4.1 Rationale for Selecting Tasmania as AgriSuit Pilot Region

Tasmania was chosen as the starting point for AgriSuit because it offers the right mix of economic importance, data quality, and regional focus needed to demonstrate the model's potential, with current state accounts and recent sector scorecards indicating agriculture's substantial contribution to the economy and continued growth momentum **tenpercentgsp**. Agriculture plays a much larger role in Tasmania's economy than in most Australian states, and the Agri-Food Scorecard shows the sector's gross value rising in 2022–23, underscoring relevance for piloting decision tools that can strengthen productivity and regional planning **howlett2025**

A key advantage of working in Tasmania is the availability of the Tasmanian Enterprise Suitability Model (TESM), which applies a rule-based most limiting factor approach and is delivered at approximately 80 m resolution suitable for farm and regional assessments. TESM delivers locally calibrated, rule-based suitability at higher resolution with explicit most-limiting-factor logic, enabling parcel-level recommendations aligned to Tasmanian conditions. TESM and related Digital Soil Assessment resources explicitly combine soil attributes with climate variables to produce categorical suitability classes, enabling transparent traceability from factor thresholds to parcel-level outcomes. Tasmania also stands out for its open access to high-quality spatial data via the LIST (Land Information System Tasmania), which provides integrated land and property information and public interfaces (e.g., LISTdata and LISTmap) used by industry and the general public **tasmania_list_2025**.

Altogether, positioning AgriSuit in Tasmania demonstrates the model under a data-rich, fine-resolution regime, thus makes Tasmania the most suitable region to pilot AgriSuit and demonstrate how localized, fine-resolution suitability mapping can transform land-use and crop planning decisions when connected to real-time property information.

4.2 The Tasmanian Enterprise Suitability Map (TESM)

The [Tasmanian Enterprise Suitability Map \(TESM\)](#) represents a mature, data-driven framework for assessing agricultural potential across Tasmania. Originating from the [Wealth from Water](#) program initiated in 2010 by the Tasmanian Department of Primary Industries, Parks, Water and Environment, which has since been restructured and renamed as the Department of Natural Resources and Environment Tasmania ([NRE Tas](#)) in collaboration with the Tasmanian Institute of Agriculture (TIA) and the University of Sydney, TESM sought to provide an objective, reproducible, and spatially explicit basis for agricultural development planning. The program was itself a direct response to the Innovation Strategy **west2009**, which identified irrigation expansion as a transformative opportunity for Tasmania's regional economy. The pilot phase initially covered about 70,000 hectares in the

Meander Valley and Midlands, subsequently expanding to state-wide coverage through initiatives such as [Water for Profit](#). In this progression, the emphasis shifted from qualitative soil surveys to Digital Soil Mapping (DSM) and Digital Soil Assessment (DSA) techniques that operationalised soil and climate data into decision-support tools for land managers and investors **kidd2020**.

TESM's analytical foundation rests upon the integration of three primary data domains, consisting of soil, climate, and topography. The domains are processed through quantitative spatial modelling. The soil component draws on more than 900 core samples describing attributes such as pH, clay content, drainage class, and stone content **kiddwebb2015b**. [Climate inputs](#) originate from a network of over 270 temperature sensors distributed across representative terrain positions, enabling fine-scale modelling of variables such as frost risk, chill hours, growing-degree days, and extreme heat frequency. Topographic variables, derived from Shuttle Radar Topography Mission (SRTM) data, gamma-radiometric imagery, and surface geology, capture slope, aspect, and elevation effects. These heterogeneous datasets are standardised and co-registered, then resampled at 80 m and modelled using regression-tree and regression-kriging approaches to produce spatially continuous soil-attribute predictions with validation accuracy appropriate for regional-scale land evaluation (Kidd & Webb, 2015). The result is a three-dimensional digital representation of the Tasmanian soil landscape that supports downstream suitability analyses.

Table 1: *List of All Variables in TESM*

Class	Variables
climate	Air temperature, Chill hours, Daily maximum temperature, Extreme heat risk, Frost risk, Growing Degree Days, Growing Season Temperature, Heat stress, Hot weather during summer, Rainfall
soil attribute	Depth to sodic layer, Duplex soil, Electrical conductivity, Exchangeable calcium, Exchangeable magnesium, Soil depth, Soil Depth, Soil drainage, Soil texture, Stone abundance
topography	Elevation, Slope

Following early operational success, the project advanced to produce state-wide 3D soil-attribute maps at high resolution, which were integrated into the national [Terrestrial Ecosystem Research Network \(TERN\)](#) and [GlobalSoilMap.net](#) repositories. Complementary climatic projections from [Climate Futures for Tasmania](#) were incorporated to generate suitability layers for 2030 and 2050 under [Representative Concentration Pathway \(RCP\) 8.5 scenarios](#). Each enterprise is therefore represented by three temporal surfaces of current (2018), near-future (2030), and long-term (2050), enabling

preliminary adaptation planning. An aggregated [Enterprise Versatility Index \(EVI\)](#) summarises the number of viable enterprises/crops per grid cell, thereby identifying land parcels capable of diversified production. The integration of these high-resolution soil and climate grids has substantially improved Tasmania’s ability to visualise spatial heterogeneity in agricultural potential.

The Tasmanian wine sector provides a salient example of TESM’s applied relevance. Building on earlier climate modelling [webb2018](#), the Enterprise suitability mapping and Tasmania’s wine sector [brullo2025](#) demonstrates how TESM outputs guide site selection for viticulture by combining frost-risk surfaces, growing-degree-day thresholds, and soil-drainage indices. These analyses have supported Tasmania’s transition toward high-value sparkling and cool-climate table wines, where irrigation expansion and land-use zoning are strategically coordinated with biophysical suitability. TESM data are increasingly used by policy agencies to outline priority irrigation districts and agricultural investment zones, reinforcing its dual role as a scientific and economic planning instrument.

Table 2: *List of All Current Crops in TESM*

Type	Crops
cereals	barley, linseed, wheat
perennial horticulture	blueberriesNHB, blueberriesSHB, cherries, hazelnuts, olives, raspberries, sparklingwg, strawberries, tablewg
vegetables	carrots, carrotseed, onions, potatoes
pharmaceuticals	hemp, poppies, pyrethrum
pastures	cocksfootcontinental, cocksfootmediterranean, lucerne, phalaris, redclover, ryegrass, strawberryclover, tallfescuecontinental, tallfescuemediterranean, whiteclover
forestry	E_globulus_tree, E_nitens_tree, P_radiata_tree

As shown in [Table 2](#), TESM catalogue comprises 33 crops grouped into seven major commodity classes. Given Tasmania’s comparatively limited arable extent, fragmented parcel structure, and frost-prone microclimates, crop choices must emphasise tolerance to wider climatic variability (e.g., frost and cool-season conditions) and the ability to sustain viable yields at smaller spatial scales than are typical on mainland Australia. Consequently, suitability assessments and investment screening prioritise climate-resilient cultivars, management flexibility (e.g., irrigation/readiness for frost mitigation), and systems that maintain productivity under tighter land and environmental constraints.

Despite its technical sophistication, the Tasmanian Enterprise Suitability Model (TESM) has several

recognised limitations. Its spatial accuracy depends on data density, declining in sparsely surveyed regions. The assumption of universal irrigation can overestimate suitability in dryland areas, and deterministic thresholds do not account for uncertainty, making results sensitive to small input changes. TESM also provides a static snapshot rather than a dynamic forecasting tool, and the administrative process for adding new crops limits responsiveness to emerging industries.

Nevertheless, TESM remains a significant contribution to digital soil assessment in Tasmania. It represents one of Australia's most comprehensive applications of digital soil modelling, integrating DSM, rule-based evaluation, and climate projections to support evidence-based agricultural planning. While its fixed thresholds and irrigation assumptions highlight the need for more flexible and uncertainty-aware systems, TESM continues to serve as a robust first-pass planning tool that translates complex soil–climate data into practical, interpretable suitability information.

4.3 Approaches to Crop Suitability Mapping

Crop-suitability frameworks vary widely across regions in data quality, model design, and agronomic priorities. For this project, linking suitability models with property listings to inform land-use investment is to understand how alternative approaches affect comparability, interpretability, and system integration. Tasmania's TESM provides a deterministic, rule-based baseline; other systems adopt probabilistic or hybrid methods that quantify uncertainty and simulate climate responses. Comparing these approaches shows that the concept of coupling suitability with market data is globally transferable, but implementation depends on each system's data structures, class vocabularies, and decision logic **malone2015**.

Strengths are transparency, auditability, and direct traceability from factor thresholds to parcel-level outcomes. Constraints include boundary brittleness near cut-offs, non-compensation among factors, and dependence on official class schemes and legends. For integration with listings, deterministic maps are straightforward to explain to non-specialists, but require careful legend normalisation across crops and explicit notes on native resolution and epochs.

Probabilistic and fuzzy extensions These quantify class likelihoods or membership grades and can propagate input uncertainty, reducing edge effects and supporting risk-aware screening. They improve decision confidence but introduce additional layers (probabilities, intervals) that must be summarised coherently for property workflows. Integration benefits from storing both the baseline categorical class and a compact uncertainty statistic (e.g., confidence band or top-two class probabilities) at parcel scale.

Hybrid and modular systems Phenology-aware or scenario-rich designs (e.g., New Zealand implementations that combine frost risk with bud-break timing) **smap2021** mitigate single-indicator

brittleness and make climate sensitivity explicit. Global frameworks (e.g., FAO GAEZ) extend scope to yield and yield-gap analyses with scenario pathways, trading local detail for coverage and comparability [fao2009](#). For real-estate linkage, hybrids are attractive because they expose both current classification and scenario response, but they demand clear schema bridges (class labels, epochs, units) and careful downscaling when parcel-level decisions are required.

Implications for linking suitability with property markets:

- **Comparability:** Class schemes, colour palettes, and granularity differ across crops and systems; cross-crop and cross-region comparisons require legend harmonisation and explicit metadata.
- **Resolution and epochs:** Native grid size and climate windows must be surfaced to avoid over-precision at parcel boundaries and to enable like-for-like comparisons over time (e.g., present vs 2030/2050).
- **Uncertainty communication:** Where probabilistic outputs exist, parcel-level summaries should report both the categorical class and a compact confidence indicator.
- **Licensing and custodianship:** Data access, update cadence, and the ability to add crops or modify rules are governed by custodians; portability to new jurisdictions requires new rule packs and validation with local experts.
- **Workflow integration:** Suitability layers must align with listing attributes (location, area filters, water access) and cadastral geometries to support due-diligence tasks without GIS expertise.

In summary, deterministic, probabilistic, and hybrid frameworks each offer distinct trade-offs in interpretability, risk communication, and scalability. The underlying idea—integrating suitability evidence with property listings to support investment decisions—is broadly applicable, but delivering it in practice requires custom schema mapping, legend normalisation, and jurisdiction-specific rule packs. The next section turns to GIS-enabled Spatial Decision Support Systems (SDSS), showing how interactive platforms operationalise these requirements through reproducible overlays, parcel-level summaries, and scenario/uncertainty toggles appropriate for end user workflows.

4.4 Geographic Information Systems (GIS) and Spatial Decision Support

Geographic Information Systems (GIS) underpin the analytical foundation of this project by enabling the integration and visualization of diverse spatial data, environmental suitability layers from the Tasmanian Enterprise Suitability Model (TESM), and property attributes from real-estate listings within a unified decision environment. In the context of this study, GIS functions not as a broad mapping tool but as a decision-support mechanism, linking agronomic intelligence with market

accessibility to inform sustainable agricultural investment. GIS facilitates multi-layer spatial analysis and delivers parcel-level insights critical to evidence-based land use decision making [sharifi1994](#).

Spatial Decision Support Systems ([SDSS](#)) extend traditional GIS capabilities by embedding rule-based and multi-criteria evaluation (MCE) models within interactive interfaces. As [carver2001](#) noted, the evolution from stand-alone GIS models to participatory, web-based SDSS frameworks enhanced transparency and stakeholder engagement in spatial planning. This same paradigm underpins the present project's design, operationalising TESM data through a web-based dashboard that allows distributed users (e.g., farmers, investors, and policymakers) to explore suitability and property information simultaneously (Rinner, 2003).

The selection of a Shiny-based GIS interface reflects both technical and strategic considerations. Shiny provides an open-source, interoperable, and reproducible environment within the R ecosystem, where the TESM datasets, statistical models, and spatial processing workflows already reside [chang2023shiny](#). This ensures seamless integration with R spatial packages such as `sf`, `terra`, and `leaflet`, supporting dynamic map rendering, spatial querying, and user-driven filtering without external proprietary dependencies. Shiny's reactive programming structure also enables real-time interaction between crop suitability layers and property attributes, allowing users to apply filters (e.g., crop type and municipality) and instantly visualise relevant parcels and listings. Shiny's accessibility and documentation facilitate rapid prototyping and deployment of interactive agricultural dashboards, as demonstrated by numerous applications internationally [appsilon2025](#).

Open-source precedents such as [Pi-VAT](#) [deval2022](#) and `ExActR` [gibbons2025](#) demonstrate that R-based, web-enabled GIS systems can effectively translate complex environmental models into accessible decision dashboards. However, the current project extends these applications beyond environmental management to the agri-real-estate domain—embedding suitability intelligence (AgriSuit) directly within property search and investment decision workflows. By operationalising TESM data through Shiny, the system transforms GIS from a back-end analytical process into a front-end, interactive decision interface, aligning with the project's objective of bridging agronomic insight and market opportunity within a transparent, web-accessible platform [keenan2019](#).

4.5 Data Storage Approach

The current AgriSuit model built persists processed datasets and analytical outputs in DuckDB, an embedded, columnar SQL engine running inside R. This configuration minimises setup overhead, supports rapid iteration with R Shiny, and retains reproducibility by saving the complete database to a single file under version control.

DuckDB has performance speed that suitable for exploratory development and demonstrations, since

it provides low-latency analytical queries (e.g., parcel-level pixel aggregations, filterable property views) and seamless input/output with CSV, enabling simple, scriptable data exchange across pipeline stages **kohn2022**.

Constraints of the prototype. DuckDB is not optimised for concurrent multi-user access, role-based security, or high availability. As usage scales, risks emerge around contention, access control, backup/restore guarantees, and operational observability **mozzillo2025**. A production instance should migrate to a stand-alone database server or managed cloud service.

4.6 Real-Estate Listing Platforms and Agricultural Land Information

Mainstream Australian real estate platforms typically emphasize transactional details such as parcel size and tenure but seldom include verified biophysical data like soil chemistry, drainage class, frost metrics, or slope **nationalcouncil2025**. When “suitable crops” are mentioned, these are largely anecdotal, seller-supplied claims with no reference to formal datasets or transparent suitability rules, requiring buyers to manually cross-reference external soil and climate sources **bhaker2025**. This fragmented process systematically disadvantages non-GIS specialists who cannot easily assemble soil, climate, and topography evidence to support investment or planning decisions **lee2023**. Studies highlight a broader lack of standardized, integrated agricultural datasets across property platforms, perpetuating risks of misinformation and reduced market transparency **dumbrell2024**. The absence of scientifically grounded suitability layers fundamentally undermines credible decision-making in agricultural land investment.

Existing literature and planning frameworks confirm that this fragmentation is systemic. **snelder2023** emphasise that land-use suitability is not an intrinsic property of a parcel but the outcome of composite indicators like soil, terrain, climate, management, and environmental objectives, weighted according to regional priorities. Without transparency in these indicators or access to underlying data, any claim of “crop suitability” remains contestable. Similarly, governance materials such as Tasmania’s Agricultural Land Mapping Project (ALMP) and the associated zoning guidelines recognise that automated desktop models can yield anomalies and must be interpreted alongside expert review and verified datasets, including the Land Potentially Suitable for Agriculture layer and water-resource portals. These sources reinforce a shared message: sound agricultural assessment depends on integrated, multi-layer spatial evidence, not static property metadata.

The key challenge hindering the integration of advanced land-suitability models with real estate listings lies in institutional and policy barriers that prevent spatial datasets from being shared or aligned across agencies and platforms **ma2024**. Critical agricultural, water, and zoning information is managed separately from property marketplaces, resulting in fragmented data flows that leave

buyers without access to verified, indicator-driven suitability evidence **adams2022**. This disconnect between technical data and transactional platforms preserves a systemic gap in market transparency and inhibits evidence-based agricultural investment decisions.

AgriSuit is designed to close this gap. By directly linking TESM-derived suitability data with live property listings in an interactive interface, it connects verified spatial intelligence with market information and demonstrates how both technical and governance barriers can be overcome in practice.

5 Data Sources and Description

This section outlines the four datasets integrated into the AgriSuit platform; property listings from Domain.com.au, crop-specific environmental suitability maps (TESM), cadastral parcel boundaries from LISTMap, and address-point data from LISTMap. Analyses were conducted in R (version 4.3.0 or above) using `rvest`, `dplyr`, `purrr`, `sf`, `terra`, `exactextractr`, and `knitr`, `kableExtra`. The interactive prototype uses `shiny` and `leaflet`. Each dataset contributes a unique spatial or semantic dimension that collectively enables parcel-level crop-suitability analysis and property–crop linkage. Table 3 summarises the provenance, file characteristics, and licensing status of each dataset.

Table 3: Data Descriptions

	Property Listing	TESM Crop Rules	LISTMap Cadastral Parcel	LISTMap Address Point
Source	Domain.com.au	LISTMap Open Data	LISTMap Open Data	LISTMap Open Data
File Type	HTML (scraped); CSV processed	GeoTIFF + .vat.dbf	SHP + DBF + PRJ (+ aux files)	SHP + DBF (+ aux files)
Metric	289 records (July 2025 snapshot)	30–80 m cell resolution	Parcel-polygon coverage (~560 k features)	~340 k address points statewide
Criteria (Filter)	Filtered for Tasmanian listings; price parsed to numeric; land area $\geq$ 10 Ha; geocoded; deduplicated by <code>full_address</code> .	Selected 33 crops with complete raster and .VAT files	29 municipal folders merged; retained active PID records	Statewide dataset linked by PID and full address string
Limitation	Price inconsistency; missing coordinates for some records	Two crops (cherries, hazelnuts) missing .VAT files	File-type fragmentation across municipalities	Address tokens parsed by space; requires string re-assembly
Value Add	Market pricing and listing attributes for agri-investment analysis	Pixel-level suitability classification per crop	Geometric boundaries for parcel-based zonal analysis	Spatial join bridge between address and parcel datasets
License	Terms of Use	CC BY 3.0 AU	CC BY 3.0 AU	CC BY 3.0 AU

Together, these datasets form a unified analytical base linking market listings with environmental suitability and cadastral geometry.

6 Methodology

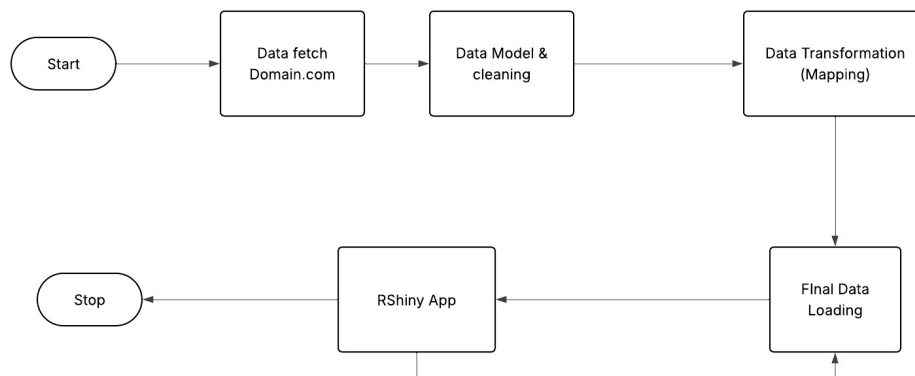


Figure 1: Project’s Flowchart Diagram

The data pipeline illustrated above Figure 1 represents a streamlined end-to-end workflow that transforms raw property listings into interactive geospatial insights. The process begins with automated data extraction from Domain.com, capturing property attributes and spatial identifiers. This raw data undergoes rigorous modelling and cleaning to ensure structural consistency and remove redundancies before being passed through a transformation stage that maps listings to cadastral parcels and crop-suitability indicators. The transformed datasets are then integrated and loaded into a consolidated DuckDB environment, forming the analytical backbone of the system. Finally, the curated data is delivered through an RShiny interface, enabling dynamic exploration, filtering, and visualisation of property-suitability relationships. Together, these stages create a cohesive data ecosystem that bridges web-sourced property data with spatial analytics, ensuring both transparency and reproducibility across the AgriSuit workflow.

6.1 Data Fetch (Domain.com.au and Supporting Datasets)

Referring to Figure 1, the work begins with the data fetch stage, where multiple datasets are collected from both online and cadastral sources to build the foundation for analysis. The primary dataset is obtained from [Domain.com.au](https://www.domain.com.au) using an automated web-scraping process. Each webpage typically displays ten property cards, and the scraper systematically iterates through all pages to extract essential details such as the property ID, full address, price, land area (in hectares), listing type (e.g., sale or lease), and corresponding URL links. Once this raw property data is gathered, it undergoes initial text preprocessing through string manipulation and pattern-matching techniques to extract structured information. For instance, numeric price values are parsed from text strings (e.g., “Price: \$1.2M”

→ 1,200,000), land area values are detected using pattern extraction, and irrelevant characters or HTML tags are removed. These operations ensure the scraped data is uniform and free from noise before further integration.

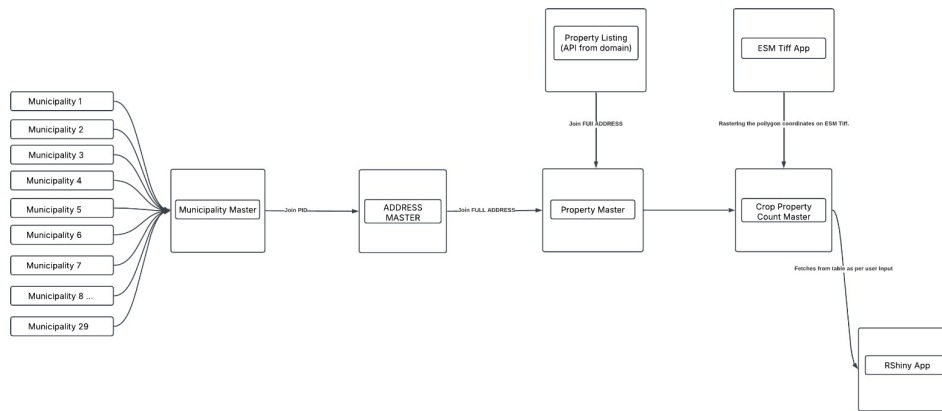


Figure 2: Data Flow Diagram

In addition to the property listing data, several related datasets are obtained from official cadastral and spatial data sources. These include the Municipality datasets, which contain cadastral parcel boundaries and polygon coordinates across Tasmania’s 29 municipalities, and the Address datasets, which provide verified address references linked to property identifiers (PIDs). These spatial and address datasets are crucial for ensuring the accuracy and georeferencing of the property information. Furthermore, the workflow incorporates Environmental Suitability Model (ESM) TIFF maps, which depict the crop suitability or feasibility for various crops across Tasmania at the pixel level. These raster layers are integral to later stages of the workflow, where they are spatially overlaid with property polygons to determine the crop suitability profile for each property.

It is important to note that the web-scraping methodology has been employed purely for the prototype phase, providing a flexible and rapid means to access publicly available data for proof-of-concept validation. However, for future scalability or commercialisation, it is strongly recommended to transition towards API-based data retrieval, leveraging official endpoints from Domain.com and cadastral data providers. An API interface would ensure higher data accuracy, stability, and compliance while reducing the operational overhead of scraping.

The cleaned and preprocessed property data is then exported to a CSV file and stored within the project workspace as an intermediate staging layer. Finally, the refined data, along with the cadastral and ESM raster datasets is loaded into the DuckDB database, forming a unified data foundation for subsequent stages such as data modelling, spatial transformation, and suitability analysis.

6.2 Data Modelling, Cleaning, and Transformation

Following the data-fetching procedures detailed in Section 6.1, the workflow transitioned into the data-modelling and integration phase, where the collected datasets were validated, cleaned, and spatially structured to form a coherent analytical foundation for the AgriSuit dashboard. This stage was essential for transforming diverse and semi-structured inputs—ranging from web-scraped property listings to official cadastral parcels—into a relationally connected database capable of supporting parcel-level analysis.

Data Cleaning and Validation

The initial step involved refining and validating the property listing and address datasets to ensure accuracy, completeness, and consistency. Raw listings retrieved from the web-scraping workflow contained missing fields, duplicated entries, and irregular formatting that could interfere with spatial joins. These records were systematically filtered, retaining only complete entries with verified location information. Textual inconsistencies such as spacing, capitalisation, and punctuation were corrected to align with Tasmanian addressing conventions. Each valid record was then geocoded to obtain latitude and longitude coordinates, ensuring positional accuracy. This process established a verified and uniform property dataset suitable for integration with official spatial layers.

Address Standardisation

After validation, the address data were standardised to ensure structural uniformity across all datasets. Individual address components—street number, street name, street type, suburb, state, and postcode, were concatenated into a composite `FULL_ADDRESS` field. This field provided a consistent textual identifier across both the web-scraped listings and the verified address-point dataset described in Section 6.1. The purpose of this step was to harmonise address representation rather than immediately establish relational dependencies, reducing ambiguity when cross-referencing between independent data sources. The outcome was a clean, standardised address framework that ensured one-to-one correspondence between listed properties and their official spatial references.

Parcel Preparation and Spatial Validation

Parallel to address standardisation, cadastral parcel layers were prepared and validated. Each municipal dataset contained standard GIS vector components, including geometry files (.shp), attribute tables (.dbf), coordinate definitions (.prj), and supporting index files (.shx, .sbn, .sbx). These files collectively define parcel boundaries and their associated attributes such as Parcel Identifier (PID) and Municipality_ID. To maintain uniformity, all datasets were re-projected into the common coordinate reference system EPSG:28355 (GDA94 / MGA Zone 55). Geometry checks ensured that parcel boundaries were topologically valid and spatially aligned with their corresponding address points. The PID field was reviewed to confirm uniqueness, thereby preventing duplication or missing identifiers.

By the conclusion of this step, the cadastral data were spatially coherent, attribute consistent, and ready to serve as the geometric foundation for integration.

Relational Database Design

With clean and verified datasets established, the workflow progressed to database modelling. A relational schema was designed to manage the interconnected property, address, and parcel data efficiently and to support reproducible spatial joins. The schema comprised five core tables: `Municipality_Master`, `Address_Master`, `Property_Listing`, `Property_Master`, and `CROP_PROPERTY_COUNT_MASTER`. The `Municipality_Master` table stored parcel geometries and associated identifiers. The `Address_Master` table contained verified addresses linked to parcels via PID, while `Property_Listing` stored the cleaned Domain.com records indexed by `FULL_ADDRESS`. The `Property_Master` table functioned as the central bridge, combining PID, `FULL_ADDRESS`, and `Municipality_ID` into unified property records. The `CROP_PROPERTY_COUNT_MASTER` table was reserved for analytical outputs generated later during the suitability computation stage. Each table was structured with explicit primary and foreign key constraints to ensure referential integrity. For instance, PID acted as the primary key within `Municipality_Master` and a foreign key within `Address_Master` and `Property_Master`. Similarly, `FULL_ADDRESS` served as the primary key in `Property_Listing` and as a foreign key reference within `Property_Master`. These relationships established the logical hierarchy connecting textual property identifiers to their spatial representations.

Join Logic and Integration

Following schema design, relational joins were performed to merge the datasets into a single queryable structure. The first join linked `Property_Listing` and `Address_Master` via `FULL_ADDRESS`, creating verified address-to-listing correspondences. The second join connected `Address_Master` to `Municipality_Master` through the PID attribute, binding each address to its cadastral parcel boundary. The resulting relational chain, `Property_Listing` → `Address_Master` → `Municipality_Master`—formed the backbone of the AgriSuit database, enabling spatially aware queries that connect market listings to physical land units. The `Property_Master` table consolidated these relationships into a unified view, providing a single record per property containing its listing details, verified address, and parcel metadata. This relational structure ensured that every property could be unambiguously traced from textual address to geospatial boundary, forming a transparent link between public market information and official land records.

System Validation and Output

Before finalising the integration, integrity tests were conducted to confirm the accuracy of all joins and key relationships. Duplicate keys, orphaned references, and inconsistent coordinate records were systematically checked and corrected. Spatial overlays were used to verify that linked parcels

accurately encompassed their associated geocoded addresses, ensuring no mismatches between tabular and geometric components. The validated output was a harmonised, fully georeferenced relational database capable of supporting both tabular and spatial queries. This structure provided a robust foundation for the analytical operations that follow in Section 6.3, where the database relationships are extended to incorporate environmental suitability data and pixel-based aggregation.

6.3 Relational Integration & Keys

The integration of property listings, cadastral parcels, address points, and crop-suitability rasters requires both relational and spatial join operations, depending on whether a shared key exists across datasets. The preferred linkage is through the canonical parcel identifier (PID), a persistent key assigned within Tasmania’s cadastral system. When this key is present in both the listing and the parcel datasets, integration proceeds as a direct relational join, thus guaranteeing one-to-one alignment and minimal spatial ambiguity.

Where such identifiers are missing, a derived key is constructed via geocoding. Each address string is first normalised (title-cased, punctuation removed, abbreviations canonicalised) and then converted to latitude & longitude coordinates using deterministic geocoding. The resulting point geometry (x, y) is spatially joined to the parcel layer through a point-in-polygon (PIP) query, assigning the parcel’s PID and attributes to the listing. This transformation converts textual addresses into spatially anchored records suitable for downstream joins with TESM raster summaries.

Address fields are preprocessed through a deterministic normalisation pipeline, applying consistent casing, token squishing, and abbreviation expansion (e.g., “Rd” into “Road”, “Tas” into “Tasmania”).

Formally, each listing address is transformed into a canonical string:

$$A_c = f(\text{street, suburb, state, postcode})$$

This ensures reproducibility in geocoding and subsequent joins with the LIST address-point dataset, where components are stored separately (delimiter by space). By concatenating these elements and appending “Australia”, the process generates a single `full_address` field used as the join key between the property listing and address-point tables. Once matched, the shared PID field provides a bridge to the parcel polygon dataset.

Multi-Stage Join Strategy

1. Listing $\longleftrightarrow$ Address Point (Textual Join). Match on normalised `full_address` with tolerance for variant spellings and spacing.
2. Address Point $\longleftrightarrow$ Parcel (Spatial Join). Using PID as a deterministic key or, when absent, point-in-polygon containment.

3. Parcel $\longleftrightarrow$ TESM Raster (Zonal Join). Overlay parcel geometries with raster layers to extract pixel-level suitability counts and compute majority-class summaries.

Each stage outputs a referentially consistent table with persistent identifiers (`listing_id`, `PID`, `crop_name`) to enable incremental refresh and audit tracking.

Integration accuracy assumes that geocoded points lie within true parcel boundaries and that the cadastral layer reflects the latest survey segmentation. Misalignments can arise from rural address ambiguity or delayed cadastral updates. For this reason, all joins record the source version and dataset date as metadata attributes:

```
record = {PID,listing_id, dataset_date, confidence, geometry}
```

This practice enables versioned reprocessing when newer cadastre or listing data are ingested.

In the case of failed deterministic joins, two complementary methods suggested below can enhance robustness:

1. Authoritative address points as intermediate keys. Many jurisdictions (including Tasmania's LIST) publish official geocoded address registers, providing a stable bridge between text-based listings and spatial polygons.
2. Probabilistic record linkage. Algorithms such as Jaro–Winkler (Jaro) or Levenshtein distance can detect near-matches in noisy vendor feeds, supporting semi-automated integration where deterministic keys are missing (Andrzejewski et.al, 2024).

In summary, the integration pipeline couples canonical identifiers with spatial containment logic, ensuring that each listing is reliably matched to a physical parcel and corresponding crop-suitability raster summary. This hybrid key architecture provides a reproducible and scalable foundation for AgriSuit's unified geospatial–market database.

6.4 Pixel Suitability

Geospatial inputs consist of layered surfaces and geometries that must be spatially consistent. A raster is a gridded surface in which each cell (pixel) stores a value for a single thematic layer (e.g., enterprise suitability); multiple layers coexist and must share a coordinate reference system to be comparable. In contrast to CSVs or single SQL tables, useful analysis emerges only after geometries (parcels, address points) and rasters are aligned and joined with stable identifiers. In our pipeline, relational keys from Section 6.3 (notably `PID` and canonical `full_address`) anchor this alignment, while spatial containment ensures that per-parcel summaries correspond exactly to what is rendered.

The overall suitability at pixel scale follows the Most-Limiting-Factor (MLF) principle, “for each crop, soil, climate, and terrain sub-rules yield categorical classes, and the strictest class governs the pixel’s final label”. This non-compensatory operator makes the mapping transparent and risk-averse, so parcel-level tallies faithfully reflect limiting conditions.

Operationally, the parcel geometry is validated and reprojected to the crop raster’s CRS, the raster is windowed to a buffered cut-out around the parcel for efficiency, and the value attribute table (VAT) is used to translate numeric codes into human-readable class names. For pixel counting, the raster is masked by the parcel polygon and intersecting pixels are retrieved with their inside-parcel fractions; a 50% majority rule excludes boundary slice and reduces double-counting at edges. Counts are aggregated by suitability class, then stamped with parcel identifiers (PID), canonical address, municipality, and crop name, yielding a per-parcel x per-crop summary that is written to the analytics table for downstream queries and filtering.

For pixel mapping, the same alignment and VAT translation steps are used, but the goal is an interpretable visualisation. The raster is clipped exactly to the parcel footprint, class labels are applied, and a stable colour scheme from the legend registry is assigned (with a neutral fallback when a class lacks a registered colour). Only classes present within the parcel are retained, producing a clean, colour-coded map tile that corresponds one-to-one with the counts stored in the summary table and the relational bridges.

6.5 Dashboard Development (AgriSuit Application)

Purpose

AgriSuit operationalises parcel-level suitability by exposing TESM-derived indicators and listing metadata through a minimal, auditable interface designed for non-specialists. The goal is reproducible, decision-ready outputs with clear provenance and bounded query scope.

Architecture

AgriSuit follows a three-layer design that separates data preparation, service orchestration, and interactive rendering.

- Data layer. TESM rasters are pre-summarised to parcel-level indicators (majority suitability class; limiting factors). Listing metadata (address, locality, indicative price) is ingested and reconciled to cadastral parcels; non-market parcels (e.g., national parks, charitable holdings, government land) are filtered upstream.
- Service layer. Parameterised queries handle geocoding/matching, apply validation rules, and stamp results with provenance (dataset version, refresh timestamp).

- UI layer (R Shiny). Reactive modules manage filters, a unified results set (table + map), and state (validation messages, row caps, legends).

Core features

A compact summary of AgriSuit's functionality is presented in Table 4.

Table 4: *Core features*

Feature	Details
Filtering	Crop Category; Crop; Class Label; Municipality; Full Address; Price Min; Price Max; Maximum Rows.
Validation	Inputs must satisfy (Crop OR Class Label) AND (Municipality OR Full Address); ambiguous or under-specified queries are rejected with prompts.
Results (Table)	Parcel ID; majority suitability class; limiting factors; listed price; direct link to the public listing.
Results (Maps)	• Context map (WMS) situating the parcel within Tasmania. • Parcel-detail map with pixel-level suitability and a crop-specific class legend.
Registration	Email (required); Full Name; Phone; at least two preferences from (Preferred Crop; Preferred Municipality; Class Label; Price Min; Price Max); Alert Frequency.

7 Results

These steps document the typical user journey, from initial view to personalised alerts.

Default Dashboard Interface Display

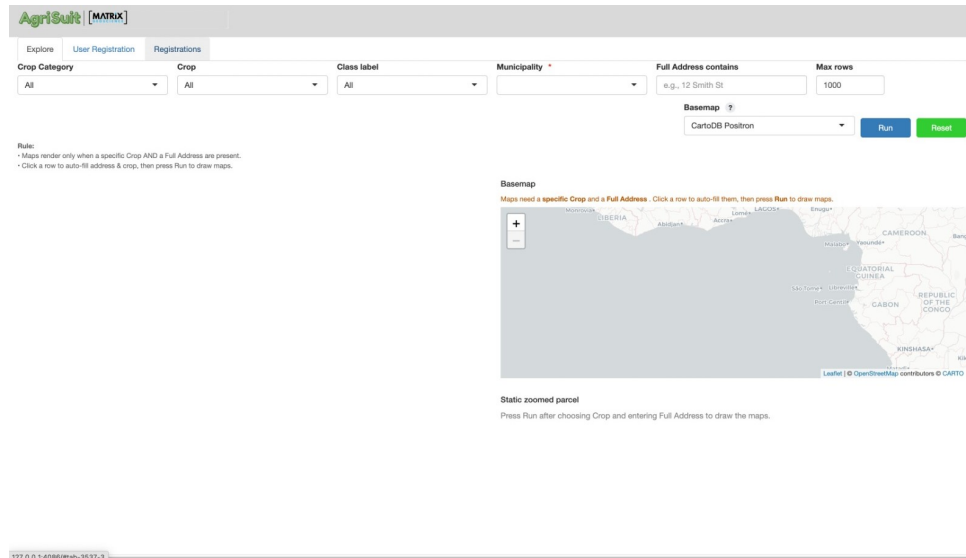


Figure 3: AgriSuit Default Display

When the application is first opened, users are presented with the default dashboard interface under the Explore tab, as seen on Figure 3. The layout displays filter controls across the top of the screen, including crop category, specific crop, suitability class, municipality, full address search, and maximum rows. At this stage, both map windows remain inactive until valid inputs are provided, ensuring users are not overwhelmed with information.

Target Preferences Selection Step (Minimum Criteria Requirement)

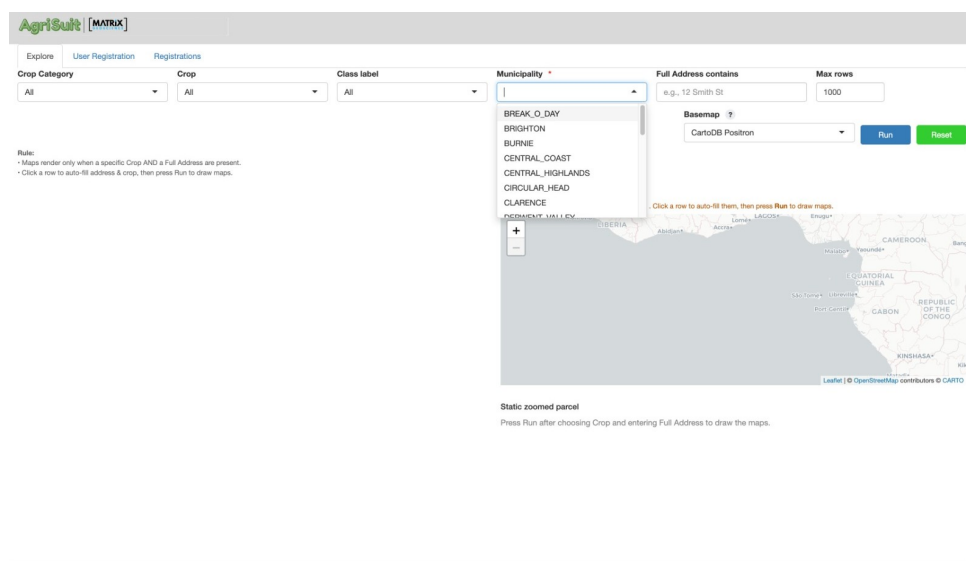


Figure 4: Preferences Selection

User begin interacting with the system by selecting their target preferences through the dropdown filters, displayed in Figure 4. To maintain clarity and reduce unnecessary processing, the dashboard enforces a minimum filter requirement: at least one of the agricultural filters (Crop or Class Label) must be selected together with a Municipality. This ensures that each search has both a crop-related constraint and a geographic focus. Without this paired selection, the dashboard will not execute a query. Additional filters, such as address keywords or suitability classes, can be applied to further refine results.

Tabular Results Display After Running Query

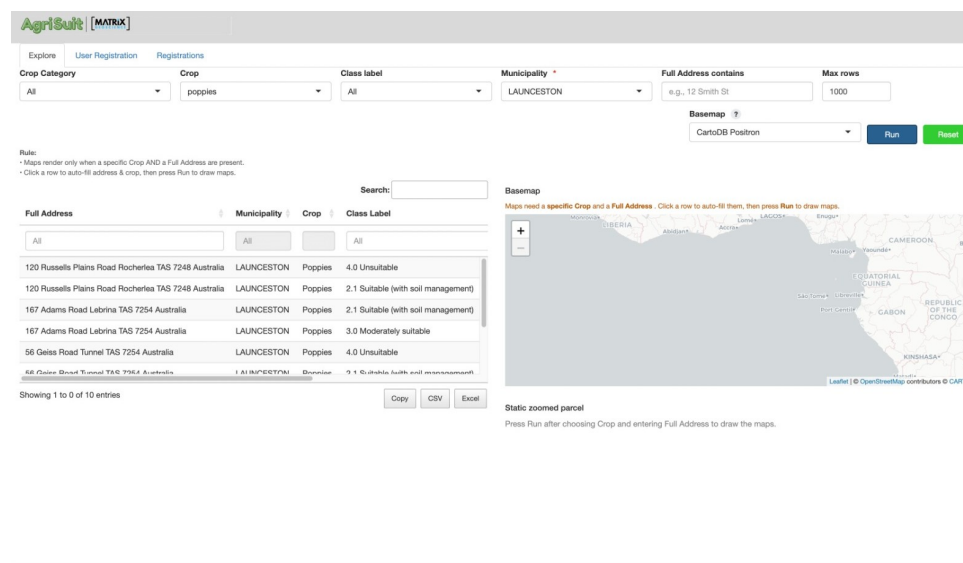


Figure 5: Table Output

After satisfying the minimum criteria and pressing the Run button, the dashboard retrieves matching properties from the back-end database. The lower-left section displays a tabular view of parcel addresses, suitability classifications, and related attributes. Built-in controls allow users to sort, search, and export results to CSV or Excel formats. Map outputs remain unchanged until a specific parcel is selected (Figure 5).

Automatic Field Population via Row Selection

When a property row is clicked, depicted in Figure 6, the system automatically populates the Full Address and Crop fields. This behaviour reduces manual typing, minimises errors, and accelerates navigation. The user can then press Run again to trigger geospatial visualisation based on the selected parcel.

Full Output Rendering (Dynamic Basemap and Static Parcel Map)

Each returned property includes a “View” control that opens the original listing in a new browser tab, as seen on Figure 7. This link-out transitions users from on-platform analysis to off-platform transaction workflows, enabling concrete next steps such as contacting the agent, arranging an inspection,

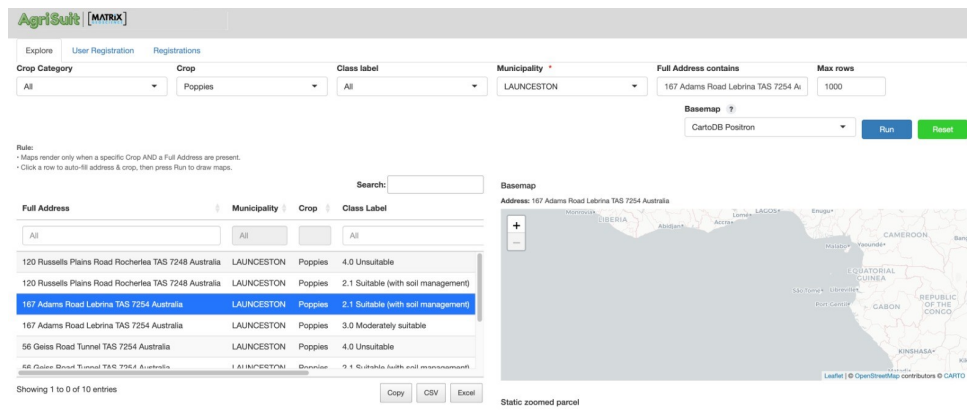


Figure 6: Row Click Feature

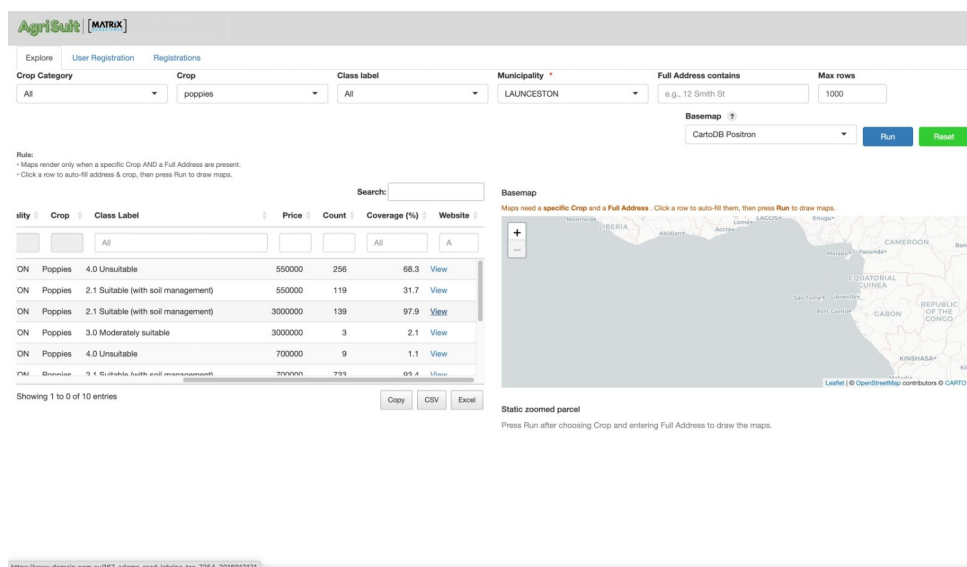


Figure 7: AgriSuit-Property Listing Hyperlink

requesting disclosure documents, verifying water access and zoning details, and proceeding with negotiation or offer submission as appropriate.

Full Output Rendering (Dynamic Basemap and Static Parcel Map)

Once a valid crop–address combination is provided, both mapping windows become active:

- The dynamic basemap situates the parcel within Tasmania’s broader geography.
- The static parcel-level suitability map displays pixel classifications according to TESM suitability categories. This dual-view, shown on Figure 8, allows users to assess suitability at both regional and local parcel scales.

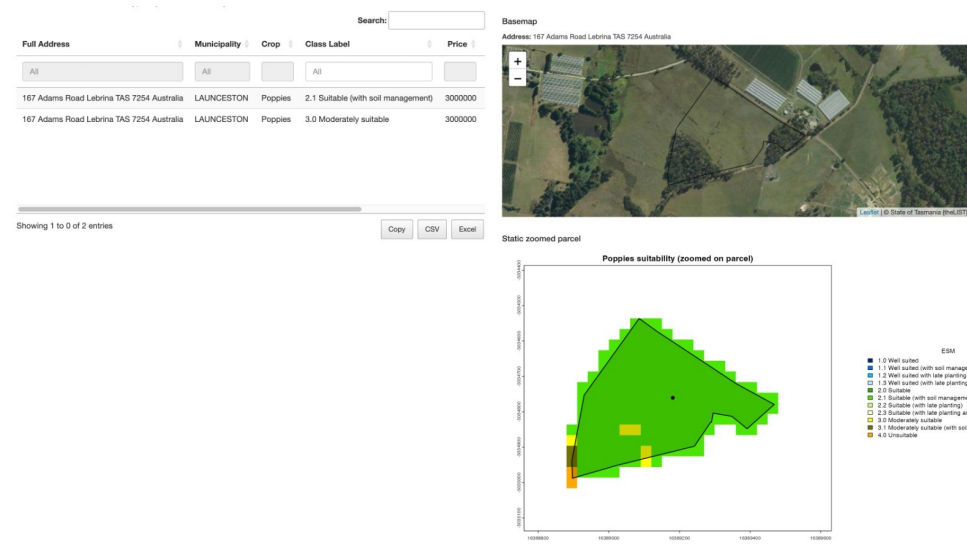


Figure 8: AgriSuit Explore Page Output

The screenshot shows the AgriSuit User Registration page. It includes a 'Subscribe for personalised notifications' section with fields for 'Email address', 'Full name (optional)', and 'Phone (optional)'. Below this are dropdown menus for 'Preferred crop', 'Preferred municipality', and 'Class label', along with input fields for 'Price min (AUS)' and 'Price max (AUS)'. There is also a section for 'Alert frequency' with a 'Weekly' dropdown and a 'Notes (optional)' text area. A 'Save registration' button is at the bottom. A 'Preview' section on the right shows the user's selected preferences.

Figure 9: User Registration Page

Personalised Alerts Through User Registration

Users seeking automated notifications can register their preferences through the User Registration tab (Figure 9). Options include preferred crop types, municipalities, suitability classes, price ranges, and alert frequency. When new properties meet these criteria, users receive personalised updates, transforming the process from manual searching to proactive discovery.

8 Discussion

8.1 Interpretation of Findings

The AgriSuit project validates that combining TESSM crop suitability intelligence with active property listings can meaningfully improve how agricultural land is evaluated in Tasmania. The system automates the linkage between property parcels and suitability classifications, delivering both numerical assessments and spatial visualisations that users can instantly interpret. Through the integration of TESSM raster layers, cadastral geometries, and property databases, the platform demonstrates that parcel-scale suitability analysis can convert static environmental datasets into practical market intelligence.

Dashboard functionality confirms that users can effectively filter properties by crop type or suitability class, with real-time mapping providing immediate geographic context. Direct links to external property listings add practical utility, allowing users to transition smoothly from analytical assessment to property investigation. This integration of environmental science and market information eliminates manual data reconciliation, accelerating decision-making for investors, planners, and developers while improving analytical reliability.

More broadly, these findings highlight how geospatial automation can address the persistent gap between environmental assessment frameworks and property technology infrastructure. The prototype establishes that publicly available datasets and open-source tools including TESSM, LISTMap, and RShiny, can be orchestrated into a scalable decision-support system without dependence on commercial platforms. The platform's validation protocols and interface design ensure query specificity, minimising ambiguity and computational overhead while preserving analytical rigour.

In summary, AgriSuit successfully operate enterprise/crop suitability data for practical investment applications. It fundamentally changes how crop potential is assessed at the property level, establishing a framework for transparent, evidence-based agricultural planning throughout Tasmania.

8.2 Strengths and Limitations

This work is presented as a prototype that demonstrates the feasibility of unifying TESSM/LISTmap suitability layers with property data in a single, auditable workflow. The current build implements reproducible polygon-raster aggregation alongside consistent CRS/grid handling and explicit re-sampling rules. An open-source stack (R Shiny, Leaflet, DuckDB) supports transparent inspection, testing, and extension, providing a sound basis for measured evaluation.

On that foundation, the integrated interface could reduce context-switching in due diligence, shorten time to insight for non-GIS users, and supply a defensible evidence trail for planning and advisory

tasks. However, such benefits must be verified in real workflows before any claims are made. Several constraints shape interpretation: outputs reflect deterministic TESM classes without uncertainty propagation; class flips can occur near rule thresholds (brittleness); parcel-level modal labels may obscure fine-scale hotspots (aggregation bias); and grid/resampling choices can affect edge pixels. Climate views are limited to available scenario epochs, external listings bring refresh/schema/ToS risks, and portability beyond Tasmanian rule libraries remains untested. A catalogue constraint also applies: the released TESM distribution exposes a fixed set of roughly 33 crop options, and adding new crops typically requires custodian updates by NRE Tas rather than an application-side change. Finally, there is legend inconsistency across crops, both in the number/naming of classes (e.g., 1.0–4.0 vs. 1.0, 1.1, 2.0, 2.1 ... 3.3) and in colour assignments, which can hinder cross-crop comparison unless explicitly normalised in the interface.

Validation will proceed by cross-checking parcel results against LISTmap samples, running boundary and area-balance QA (including $\geq 50\%$ coverage checks), and attributing variance through an explicit error budget for resampling, parcel geometry, and geocoding. Small user pilots will then measure task time, SUS, and qualitative pain points before any efficiency claims are reported.

8.3 Future Scope

A practical next step is to make suitability analysis scenario-aware. Integrating the TESM 2030 and 2050 layers as first-class toggles that are clearly labelled by epoch, native resolution, and any re-sampling would enable like-for-like comparisons between present and projected suitability at parcel and regional scales without altering the baseline logic. Complementing this, uncertainty overlays reported alongside TESM classes to adhere risk-aware screening; Monte-Carlo or fuzzy membership summaries at pixel and parcel levels would expose confidence and boundary sensitivity while preserving the official categorical outputs for compliance and communication.

Broadening scope from “can it grow here?” to “is it viable here?” will require additional inputs that remain auditable. Two priorities stand out: hydrology and water-access indicators (distance to secure water sources, scheme connection, allocation/permit status) and economic layers (yield proxies or observed yields where available, logistics distance to processors/ports, and indicative price/cost ranges). Together, these additions would support a transparent suitability-plus-viability view aligned with investment decisions.

Operational maturity likewise depends on dependable data operations. Automated refresh and synchronisation of listings and cadastral layers, provenance/version tags that bind map releases to app builds, and service decomposition, an API endpoints with cloud-optimised GeoTIFF tiling and caching to help sustain freshness and performance. A personalised notification module (filters for crop,

municipality, class, price) can then shift the tool from pull-only exploration to timely, criteria-driven alerts without altering the analytical logic.

Deployment path (multi-user and secure). A production instance should migrate to a stand-alone database server or managed cloud service, such as PostgreSQL with PostGIS on AWS RDS, Google Cloud SQL, or Azure Database for PostgreSQL. This stack supports spatial types and indexes (PostGIS), automated backups, point-in-time recovery, encryption at rest/in transit, monitoring/alerting, and elastic compute/storage to meet peak demand.

There are also operational considerations, a cloud database enables controlled capacity scaling (up/down with load), isolates analytical jobs from the web tier, and formalises data governance (least-privilege access, audit logs, scheduled maintenance windows). The migration plan should also pin data provenance (schema versions, release tags) and formalise refresh jobs for listings and cadastral layers.

The same concept is portable to other states, provided a custom setup is undertaken: jurisdiction-specific rule packs and class vocabularies, data adapters for local soil/climate rasters and CRS/metadata conventions, appropriate licensing and attribution, expert validation against local practice, and performance tuning for differing data volumes and service constraints.

A pragmatic roadmap follows this sequence: deliver climatic scenario toggles and labelling; add uncertainty overlays with concise summary statistics; introduce water-access and core economic indicators; harden data refresh, provenance, and notifications; execute the database migration; then package a jurisdiction template (rule-pack plus data-adapter kit) for pilots beyond Tasmania.

9 Conclusion

AgriSuit successfully demonstrates that agricultural suitability intelligence can be operationalised for real-world land investment decisions. By integrating the Tasmanian Enterprise Suitability Map with live property listings, the system transforms complex geospatial data into accessible, actionable insights. The prototype automates parcel-level suitability assessments and connects users directly to property markets, proving that environmental datasets can meaningfully inform commercial decision-making.

The project achieves its core objective of bridging research-grade environmental data with practical property evaluation. Through the integration of cadastral boundaries, address registries, and TESM raster classifications, AgriSuit enables investors to identify agriculturally viable land and immediately access detailed property information. This functionality reduces the friction between suitability analysis and investment action, offering genuine value to potential users.

Beyond its immediate application, AgriSuit illustrates how open government datasets can be repurposed into decision-support infrastructure. The platform provides a working model for embedding environmental intelligence into PropTech systems, with implications for agricultural planning, sustainable land-use policy, and evidence-based investment frameworks. By making suitability data transparent and accessible, the system supports Tasmania's broader goals of informed land allocation and sustainable agricultural development.

The prototype's limitations (reliance on deterministic TESM classifications, potential inconsistencies in web-scraped property data, and computational constraints during large-scale queries) are inherent to proof-of-concept development rather than fundamental design weaknesses. These issues establish clear priorities for future development: incorporating climate projection data for 2030 and 2050, automating database updates to maintain currency, and implementing user notification systems to enhance engagement.

AgriSuit delivers a functional foundation for integrating geospatial analytics with real-estate platforms, demonstrating both technical feasibility and practical value. While modest in scope, the project validates an approach to agricultural land assessment that could be refined, scaled, and adapted to other regions and agricultural contexts. It represents a concrete step toward data-driven agricultural planning and investment decision-making in Tasmania.

10 Github Repository

The complete codebase for the AgriSuit project is hosted on GitHub and can be accessed at the following [link](#)

% Print the bibliography here (and add it to ToC)

% Prevent any later, template-inserted bibliography from printing again \let__origprintbibliography%

A Appendix

Code snippet for data wrangling (cleaning raw price to numeric column).

```
# Convert price text to numeric dollars where possible
parse_price_numeric <- function(price_text) {
  if (is.na(price_text) || str_trim(price_text) == "") return(NA_real_)

  # Normalise text: uppercase, remove commas and non-breaking spaces
  txt <- stringr::str_to_upper(price_text)
  txt <- stringr::str_replace_all(txt, "[,\\u00A0]", "")

  # Skip text-only placeholders
  if (stringr::str_detect(txt,
                          "EOI|EXPRESSION|CONTACT|TENDER|AUCTION|BY
                          NEGOTIATION|PRICE ON APPLICATION|POA|BY OFFER"))
    return(NA_real_)

  # Flexible match: find first number anywhere, with optional $
  # and optional K/M suffix
  m <- stringr::str_match(txt,
                          ".*?\\$?\\s*([0-9]+(?:\\.[0-9]+)?)\\s*([KM])?.*")

  # Only the NUMBER capture is required; suffix may be NA
  if (is.na(m[,2])) return(NA_real_)

  val <- as.numeric(m[,2])
  suf <- m[,3]

  if (!is.na(suf)) {
    if (suf == "K") val <- val * 1e3
    if (suf == "M") val <- val * 1e6
  }

  as.numeric(round(val))
}
```


Establishing the database connections (Duck Db).

```
# Creating / opening a database file
con <- dbConnect(duckdb(), dbdir = "TESM.duckdb", read_only = FALSE)
```

Consolidating the land informations into one table.

```
/* creating the final master table by joining the property listing data with
cadstral address master table.*/
```

```
CREATE OR REPLACE TABLE PROPERTY_MASTER AS
SELECT
  p.PID,
  p.POT_PID,
  p.MUNICIPALITY,
  d.full_address as FULL_ADDRESS,
  d.price_numeric as PRICE,
  ST_AsText(ST_GeomFromWKB(p.parcel_geometry)) AS PARCEL_WKT,
  ST_AsText(ST_GeomFromWKB(p.address_geometry)) AS ADDRESS_WKT,
  d.Listing_URL as Domain_URL
FROM PROPERTY_LISTING d
JOIN CADASTRAL_ADDRESS_MASTER p
  ON upper(trim(replace(d.full_address, chr(160), ' '))) =
    upper(trim(replace(p.full_address, chr(160), ' '))) where POT_PID!=0;
```

Code snippet for Pixel counting.

```
vat_lu_dedup <- dplyr::distinct(vat_lu, code, label)

out <- cell_df |>
  # Keep majority-covering cells and non-missing values
  dplyr::filter(!is.na(value), coverage_fraction >= majority_thr) |>
  # Count cells by class value
  dplyr::count(value, name = "COUNT") |>
  # Join human-readable class labels
  dplyr::left_join(vat_lu_dedup, by = c("value" = "code")) |>
  dplyr::mutate(
```

```
CLASS_LABEL = dplyr::coalesce(label, as.character(value)),
CROP        = tools::file_path_sans_ext(basename(tif_path))
) |>
# Final tidy columns (carry-through identifiers and geometry WKT)
dplyr::transmute(
  PID        = prop_row$PID,
  POT_PID    = prop_row$POT_PID,
  MUNICIPALITY = prop_row$MUNICIPALITY,
  FULL_ADDRESS = prop_row$full_address,
  Domain_URL  = prop_row$Domain_URL,
  COUNT,
  CLASS_LABEL,
  CROP,
  GEOMETRY    = prop_row$parcel_wkt,
  ADDRESS_WKT = prop_row$address_wkt
) |>
dplyr::arrange(dplyr::desc(COUNT))
```

Code for the pixel level mapping on Rshiny dashboard.

```
tif_path <- file.path(esm_dir, paste0(crop_nm, ".tif"))
validate(need(file.exists(tif_path),
               sprintf("Missing GeoTIFF: %s", tif_path)))
r <- terra::rast(tif_path)

bb <- st_bbox(st_transform(parcel_sf, st_crs(r)))
ext_zoom <- terra::ext(as.numeric(bb["xmin"]) - zoom_buffer_m,
                      as.numeric(bb["xmax"]) + zoom_buffer_m,
                      as.numeric(bb["ymin"]) - zoom_buffer_m,
                      as.numeric(bb["ymax"]) + zoom_buffer_m)
r_cut <- terra::crop(r, ext_zoom)

vat_path <- paste0(tif_path, ".vat.dbf")
cat_tbl <- NULL
if (file.exists(vat_path)) {
```

```
vat <- foreign::read.dbf(vat_path, as.is = TRUE)
vcol <- grep("^value$", names(vat), ignore.case = TRUE, value = TRUE)
if (length(vcol) == 1L) {
  lcol <- grep("(suitab|class|label|cat|legend|desc|category)", names(vat),
              ignore.case = TRUE, value = TRUE)
  if (!length(lcol)) lcol <- vcol
  cat_tbl <- data.frame(ID = vat[[vcol]], label =
                      trimws(as.character(vat[[lcol]])))
  r_cut <- as.factor(r_cut)
  levels(r_cut) <- data.frame(ID = cat_tbl$ID, class = cat_tbl$label)
}
}

pal <- get_crop_palette(crop_nm)
classes <- pal$class_label
cols <- pal$color_hex

canon2 <- function(s) { s <- gsub("\\s+", " ", trimws(s));
s <- gsub("soil mgmt", "soil management", s, ignore.case=TRUE); tolower(s) }

col_map <- NULL
if (!is.null(cat_tbl)) {
  lut_lab2col <- setNames(cols, canon2(classes))
  cm <- lut_lab2col[canon2(cat_tbl$label)]
  cm[is.na(cm)] <- "lightgray"
  names(cm) <- as.character(cat_tbl$ID)
  col_map <- cm
}

parcel_r <- st_transform(parcel_sf, st_crs(r_cut))

r_mask <- try(terra::mask(r_cut, terra::vect(parcel_r)), silent = TRUE)
if (inherits(r_mask, "try-error")) r_mask <- r_cut
r_fac <- as.factor(r_mask)
```

```
lev_df <- levels(r_fac)[[1]]
ids_in_levels
<- if (!is.null(lev_df) && "ID" %in% names(lev_df)) as.character(lev_df$ID)
else character(0)

cols_for_levels <- rep("lightgray", length(ids_in_levels))
if (!is.null(col_map) && length(ids_in_levels)) {
  hitc <- col_map[ids_in_levels]
  cols_for_levels[!is.na(hitc)] <- hitc[!is.na(hitc)]
}
```

Rshiny server code where the Rshiny user input fetches data from the duckdb database.

```
# ---- Validation (Explore) ----
has_loc_filter<- reactive({ is_present(input$municipality)
  || is_present(input$addr) })
has_target_filter <- reactive({ selected_not_all(input$crop)
  || selected_not_all(input$class) })

# ---- Build SQL (Explore) ----
build_sql <- reactive({
  req(input$limit)
  where <- c(); params <- list()

  if (selected_not_all(input$crop)) {
    where <- c(where, sprintf("UPPER(%s)=UPPER(?crop)",
                                dbQuoteIdentifier(con,"CROP")));
    params$crop <- input$crop
  } else if (!is.null(input$crop_category) && input$crop_category != "All") {
    where <- c(where,
      sprintf("UPPER(%s) IN
        (SELECT UPPER(CROP_NAME) FROM %s WHERE UPPER(CROP_CATEGORY)=UPPER(?cat))",
          dbQuoteIdentifier(con,"CROP"),
          dbQuoteIdentifier(con, CROP_MASTER)))
    params$cat <- input$crop_category
  }
```

```
}  
  
if (!is.null(input$class) && input$class != "All") {  
  where <-  
    c(where, sprintf("UPPER(%s)=UPPER(?class_label)",  
                      dbQuoteIdentifier(con,"CLASS_LABEL")));  
  params$class_label <- input$class  
}  
  
if (is_present(input$municipality)) {  
  where <-  
    c(where, sprintf("UPPER(%s)=UPPER(?municipality)",  
                      dbQuoteIdentifier(con,"MUNICIPALITY")));  
  params$municipality <- input$municipality  
}  
  
if (is_present(input$addr)) {  
  where <-  
    c(where, sprintf("UPPER(%s) ILIKE '%%'  
                      || UPPER(?addr) || '%%'",  
                      dbQuoteIdentifier(con,"FULL_ADDRESS")));  
  params$addr <- input$addr  
}  
  
where_sql <-  
  if (length(where)) paste("WHERE", paste(where, collapse=" AND "))  
  else ""  
  
select_cols <-  
  paste(dbQuoteIdentifier(con, c("FULL_ADDRESS","MUNICIPALITY",  
                                "CROP","CLASS_LABEL","PRICE","COUNT","COVERAGE","WEBSITE")),  
        collapse=", ")  
  
sql <- sprintf("SELECT %s FROM %s %s ORDER BY FULL_ADDRESS LIMIT %d",  
               select_cols,  
               dbQuoteIdentifier(con, MAIN_TABLE),  
               where_sql, as.integer(input$limit))  
  
list(sql=sql, params=params)  
})
```

```
output$sql_preview <- renderText({ build_sql()$sql })

# ---- Run query (Explore) ----
data_reactive <- eventReactive(input$run, {
  validate(
    need(has_loc_filter(),
      "Please select a Municipality or enter
      a Full Address before running."),
    need(has_target_filter(),
      "Please select either a specific Crop or a Class label
      (Crop Category alone isn't enough).")
  )
  b <- build_sql()
  safe_query(con, b$sql, b$params)
}, ignoreInit = TRUE)
```